Abdullah Sameh

DEVOPS ENGINEER

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SUMMARY

Data Science student with strong hands-on DevOps experience spanning Kubernetes orchestration, GitOps delivery, cloud infrastructure automation, and full-stack containerization. Built and deployed production-grade systems including a live federated social media platform on Kubernetes. Focused on cloud-agnostic, maintainable architectures and automation-first practices.

TECHNICAL SKILLS

Cloud	AWS (EC2, S3, IAM, VPC, EKS, CloudFront, CodeDeploy)	**Observability**	Prometheus, Grafana, Loki, Tempo, OpenTelemetry
Containers & Orchestration	Docker, Kubernetes, Helm, Kustomize	**IaC**	Terraform, Packer
CI/CD & GitOps	GitHub Actions, ArgoCD, CodeDeploy	**Scripting**	Bash, Python

PROJECTS

Mastodon — Production Kubernetes Deployment (Federated Social Media Platform) github.com/mastodon-DevOps-version

- Deployed a live federated social media platform on Kubernetes with separate microservice deployments for **web**, **streaming**, and **Sidekiq** background workers.
- Self-hosted PostgreSQL using **CloudNativePG** with a primary + replica setup for high availability, avoiding any dependency on managed DB services.
- Abstracted the storage layer with **Longhorn CSI**, making the entire deployment cloud-agnostic and decoupled from any specific cloud provider's block storage.
- Cost-optimized node architecture: stateless app tier on **Spot instances**; stateful data services (PostgreSQL, MinIO, monitoring) on **On-Demand nodes**.
- Built two independent CI/CD pipelines via **GitHub Actions** — one for the app, one for infrastructure — both synced through **ArgoCD GitOps**. TLS via cert-manager, secrets via Sealed Secrets.

MERN E-Commerce — AWS 3-Tier Production Architecture github.com/Passw0rd404/mern-ecommerce-DevOps-version

- Deployed a full-stack MERN app on AWS using a 3-tier architecture: **CloudFront** → **NLB** → **EC2 Auto Scaling Group** in a private subnet, with **ElastiCache** and **MongoDB Atlas** via PrivateLink.
- Two separate pipelines: frontend via S3 + CloudFront invalidation; backend via **CodeDeploy** blue/green. Infrastructure provisioned with **Terraform**, AMIs baked with **Packer**.
- Secured with OIDC keyless AWS auth, IaC scanning (TFLint, Terrascan, Checkov), and code quality gates (ESLint, Semgrep, OpenSSF Scorecard).

HiveBox — IoT Temperature API (Built from Scratch) github.com/Passw0rd404/HiveBox

- Wrote a **FastAPI** application from scratch aggregating real-time temperature data from global IoT sensors (OpenSenseMap API); containerized and deployed locally on **kind**.
- Integrated **Valkey** caching, Prometheus metrics, and a Grafana dashboard — full local observability stack without any cloud dependency.

HealthAI Labs — AI-Powered Health Platform (Built from Scratch) github.com/co-op-projects/HealthAI-Labs

- Built an AI health application from scratch integrating LLM APIs; containerized with **Docker Compose** — first experience shipping an AI-integrated service end to end.
- Practiced multi-service orchestration, environment variable management, and team collaboration with Git branching and pull request workflows.

EDUCATION

Bachelor's in Data Science, Faculty of Artificial Intelligence at Menoufia University Expected 2028

Minoufia University — Egypt · GPA: 3.01

CERTIFICATIONS

AWS DevOps — 90%	Amazon Web Services	View credential ↗
McKinsey Forward Program	McKinsey & Company	View credential ↗